

CONCORDIA UNIVERSITY · DEPARTMENT OF MECHANICAL & INDUSTRIAL ENGINEERING

ENGR 213 · Chapter 2 Assigned Textbook Solutions

Dennis G. Zill, *Advanced Engineering Mathematics* (7th Edition) · Midterm 1 Core Curriculum

53 Assigned Problems

Sections 2.1, 2.2, 2.3, 2.4, 2.5, 2.7, 2.8

Dual Edition Mapped (7th & 6th Ed)

100% Book Answer Verified

SECTION 2.1 · DIRECTION FIELDS & QUALITATIVE ANALYSIS

Problems 3, 4, 26, 27

PROBLEM 2.1.3 · Direction Field & Integral Curves

DIRECTION FIELDS

[QUESTION STATEMENT] Reproduce the given computer-generated direction field for $\frac{dy}{dx} = -\frac{x}{y}$. Then sketch, by hand, an approximate solution curve that passes through each of the indicated points: (a) $y(0) = 0$, (b) $y(-1) = 0$, (c) $y(2) = 2$, (d) $y(0) = -4$.

[1] Analytical Verification: Rewriting in differential form gives $x dx + y dy = 0$. Integrating both sides yields $\frac{1}{2}x^2 + \frac{1}{2}y^2 = c_0$, or $x^2 + y^2 = r^2$. The one-parameter family of integral curves consists of concentric circles centered at the origin $(0, 0)$.

[2] Slope & Isocline Analysis: The lineal element slope is $m = -x/y$. Along the y -axis ($x = 0, y \neq 0$), slopes are horizontal ($m = 0$). Along the x -axis ($y = 0$), the denominator is zero, meaning tangents are strictly vertical ($m \rightarrow \pm\infty$). The circle slopes are everywhere perpendicular to radial lines from the origin ($m_{\text{radial}} = y/x \Rightarrow m_{\text{tan}} \cdot m_{\text{radial}} = -1$).

[3] Evaluation at Given Points:

- (a) $(0, 0)$: The origin is a singular point where $dy/dx = 0/0$ is undefined. No smooth non-trivial solution curve passes through $(0, 0)$.
- (b) $(-1, 0)$: Radius $r^2 = (-1)^2 + 0^2 = 1 \Rightarrow x^2 + y^2 = 1$. The curve is the semicircle $y = \sqrt{1 - x^2}$ (or $y = -\sqrt{1 - x^2}$) with vertical tangents at $x = \pm 1$.
- (c) $(2, 2)$: Radius $r^2 = 2^2 + 2^2 = 8 \Rightarrow x^2 + y^2 = 8$. Explicit upper branch: $y = \sqrt{8 - x^2}$ defined on $(-\sqrt{8}, \sqrt{8})$.
- (d) $(0, -4)$: Radius $r^2 = 0^2 + (-4)^2 = 16 \Rightarrow x^2 + y^2 = 16$. Explicit lower branch: $y = -\sqrt{16 - x^2}$ defined on $(-4, 4)$.

Concordia Exam Pitfall: Do not write the full circle $x^2 + y^2 = r^2$ as an explicit function $y(x)$. A single solution to a differential equation must be a function; vertical tangents at $y = 0$ bound the interval of definition $I = (-r, r)$.

Official Book Answer: Solution curves are circular arcs centered at $(0, 0)$. For (b) $x^2 + y^2 = 1$; for (c) $x^2 + y^2 = 8$; for (d) $x^2 + y^2 = 16$. Point (a) is a singular point where slopes are undefined. [VERIFIED · ZILL SM]

PROBLEM 2.1.4 · Hyperbolic Direction Fields

INTEGRAL CURVES

[QUESTION STATEMENT] Reproduce the given computer-generated direction field for $\frac{dy}{dx} = \frac{x}{y}$. Then sketch an approximate solution curve through each indicated point: (a) $y(0) = 1$, (b) $y(1) = 0$, (c) $y(3) = 3$, (d) $y(0) = -5/2$.

[1] Analytical Form: Separating variables gives $y dy = x dx \Rightarrow y^2 - x^2 = c$. The solution curves are a family of rectangular hyperbolas with asymptotes $y = \pm x$.

[2] Evaluation at Points:

- (a) $(0, 1)$: $c = 1^2 - 0^2 = 1 \Rightarrow y^2 - x^2 = 1$. This is the upper branch $y = \sqrt{x^2 + 1}$, smooth on $(-\infty, \infty)$ with horizontal tangent at $(0, 1)$.
- (b) $(1, 0)$: $c = 0^2 - 1^2 = -1 \Rightarrow x^2 - y^2 = 1$. The point $(1, 0)$ is a vertex with vertical tangent; upper branch is $y = \sqrt{x^2 - 1}$ for $x \geq 1$.
- (c) $(3, 3)$: $c = 3^2 - 3^2 = 0 \Rightarrow y^2 = x^2 \Rightarrow y = x$ (the asymptote ray passing through the first quadrant).
- (d) $(0, -5/2)$: $c = (-5/2)^2 - 0 = 25/4 \Rightarrow y^2 - x^2 = 25/4$. Explicit lower branch: $y = -\sqrt{x^2 + 25/4}$, defined on $(-\infty, \infty)$.

Official Book Answer: Solution curves are hyperbolas $y^2 - x^2 = c$. (a) Upper branch $y = \sqrt{x^2 + 1}$; (b) Right branch $x^2 - y^2 = 1$; (c) Asymptote $y = x$; (d) Lower branch $y = -\sqrt{x^2 + 25/4}$. [VERIFIED · ZILL SM]

SECTION 2.1 · AUTONOMOUS EQUATIONS, CRITICAL POINTS & STABILITY

Problems 26 & 27 (Dual Edition Coverage)

PROBLEM 2.1.26 · Autonomous Rate Law & Phase Portrait

AUTONOMOUS DE

[7th Edition Problem 26 / 6th Edition Problem 42: Second-Order Chemical Reaction]

[QUESTION STATEMENT] Consider the autonomous differential equation $\frac{dX}{dt} = k(\alpha - X)(\beta - X)$, where $k > 0$ and $\beta > \alpha > 0$. **(a)** Find the critical points and construct a 1D phase portrait. Determine the limiting behavior of $X(t)$ as $t \rightarrow \infty$ for initial states $X(0) = X_0$. **(b)** Discuss the case when $\alpha = \beta$. **(c)** When $k = 1$ and $\alpha = \beta$, solve $\frac{dX}{dt} = (\alpha - X)^2$ explicitly for $X(0) = \alpha/2$ and $X(0) = 2\alpha$.

[1] Critical Points & Phase Line (Part a): Setting $f(X) = k(\alpha - X)(\beta - X) = 0$ yields two critical points: $X = \alpha$ and $X = \beta$ (with $\alpha < \beta$).

- For $X < \alpha$: Both $(\alpha - X) > 0$ and $(\beta - X) > 0 \implies dX/dt > 0$ (arrows point UP toward α).
- For $\alpha < X < \beta$: $(\alpha - X) < 0$ and $(\beta - X) > 0 \implies dX/dt < 0$ (arrows point DOWN toward α).
- For $X > \beta$: Both factors negative $\implies dX/dt > 0$ (arrows point UP away from β).

Classification: $X = \alpha$ is an **attractor (asymptotically stable)**; $X = \beta$ is a **repeller (unstable)**. Therefore, if $X_0 < \beta$, $X(t) \rightarrow \alpha$ as $t \rightarrow \infty$. If $X_0 > \beta$, $X(t)$ increases without bound.

[2] Degenerate Case $\alpha = \beta$ (Part b): When $\alpha = \beta$, the equation becomes $\frac{dX}{dt} = k(\alpha - X)^2 \geq 0$. Here, $dX/dt > 0$ for all $X \neq \alpha$. The single critical point $X = \alpha$ is **semi-stable**. If $X_0 < \alpha$, $X(t) \rightarrow \alpha$ as $t \rightarrow \infty$. If $X_0 > \alpha$, $X(t)$ increases and blows up in finite time.

[3] Explicit Separation (Part c): For $k = 1$, $\alpha = \beta$: $\frac{dX}{(\alpha - X)^2} = dt \implies \frac{1}{\alpha - X} = t + c \implies X(t) = \alpha - \frac{1}{t+c}$.

- For $X(0) = \alpha/2$: $\alpha - \frac{1}{c} = \frac{\alpha}{2} \implies c = \frac{2}{\alpha} \implies X(t) = \alpha - \frac{1}{t+2/\alpha}$. As $t \rightarrow \infty$, $X(t) \rightarrow \alpha$.
- For $X(0) = 2\alpha$: $\alpha - \frac{1}{c} = 2\alpha \implies c = -\frac{1}{\alpha} \implies X(t) = \alpha - \frac{1}{t-1/\alpha}$. As $t \rightarrow (1/\alpha)^-$, $X(t) \rightarrow +\infty$ (finite-time blowup).

6th Edition Problem 26 Alternate: $\frac{dy}{dx} = 10 + 3y - y^2 = (5 - y)(2 + y) = 0 \implies$ Critical points $y = -2$ (unstable) and $y = 5$ (asymptotically stable).

Official Book Answer: (a) Critical points $X = \alpha$ (asymptotically stable) and $X = \beta$ (unstable). For $X_0 < \beta$, $X(t) \rightarrow \alpha$ as $t \rightarrow \infty$. (b) For $\alpha = \beta$, $X = \alpha$ is semi-stable. (c) $X(t) = \alpha - \frac{1}{t+2/\alpha}$ (approaches α); and $X(t) = \alpha - \frac{1}{t-1/\alpha}$ (blows up at $t = 1/\alpha$).

[VERIFIED · ZILL SM]

PROBLEM 2.1.27 · Parametric Stability & Critical Point Analysis

PHASE PORTRAIT

[7th Edition Problem 27 / 6th Edition Problem 21: Autonomous Equilibrium Stability]

[QUESTION STATEMENT] Find the critical points and phase portrait of the autonomous differential equation $\frac{dy}{dx} = y(c - y)$ where c is a real constant. Classify the stability of each critical point for $c > 0$, $c = 0$, and $c < 0$.

(Cross-reference: 6th Ed #21/27: Find critical points and classify for $\frac{dy}{dx} = y^2(4 - y^2)$.)

[1] 7th Edition Parametric Analysis: Setting $f(y) = y(c - y) = 0$ gives critical points $y = 0$ and $y = c$.

- If $c > 0$: For $y > c$, $f(y) < 0$; for $0 < y < c$, $f(y) > 0$; for $y < 0$, $f(y) < 0$. Thus, $y = c$ is **asymptotically stable** and $y = 0$ is **unstable**.
- If $c < 0$: The ordering flips ($c < 0$). $y = 0$ is **asymptotically stable** and $y = c$ is **unstable**.
- If $c = 0$: $\frac{dy}{dx} = -y^2 \leq 0$. Slopes are negative for all $y \neq 0$. $y = 0$ is **semi-stable**.

[2] 6th Edition Polynomial Equation ($\frac{dy}{dx} = y^2(4 - y^2)$):

- Factoring: $f(y) = y^2(2 - y)(2 + y) = 0 \implies$ Three critical points: $y = -2, 0, 2$.
- Sign test on intervals: $y > 2 \implies f(y) < 0$ (decreasing toward 2); $0 < y < 2 \implies f(y) > 0$ (increasing toward 2); $-2 < y < 0 \implies f(y) > 0$ (increasing toward 0); $y < -2 \implies f(y) < 0$ (decreasing away from -2).
- **Classification:** $y = 2$ is **asymptotically stable (attractor)**; $y = 0$ is **semi-stable** (attracting from below, repelling from above); $y = -2$ is **unstable (repeller)**.

Official Book Answer: 7th Ed: Critical points $y = 0$ and $y = c$. For $c > 0$, $y = c$ is asymptotically stable, $y = 0$ is unstable. 6th Ed: Critical points are $y = -2$ (unstable), $y = 0$ (semi-stable), and $y = 2$ (asymptotically stable).

[VERIFIED · ZILL SM]

SECTION 2.2 · SEPARABLE DIFFERENTIAL EQUATIONS

Problems 1, 7, 9, 13

PROBLEM 2.2.1 · Direct Integration

SEPARABLE

[QUESTION STATEMENT] Solve the given differential equation by separation of variables: $\frac{dy}{dx} = \sin 5x$.

[1] Separation of Variables: Multiply both sides by the differential dx : $dy = \sin(5x) dx$.

[2] Direct Integration: Integrating both sides yields:

$$\int dy = \int \sin(5x) dx \implies y = -\frac{1}{5} \cos(5x) + c$$

Concordia Exam Tip: Never forget the integration constant c in indefinite integrals. For first-order equations, the constant appears immediately upon integrating.

Official Book Answer: $y = -\frac{1}{5} \cos(5x) + c$

[VERIFIED · ZILL SM]

PROBLEM 2.2.7 · Exponential Separation

SEPARABLE

[QUESTION STATEMENT] Solve the given differential equation by separation of variables: $\frac{dy}{dx} = e^{3x+2y}$.

[1] Algebraic Factoring: Using exponent rules, rewrite the right-hand side: $\frac{dy}{dx} = e^{3x} \cdot e^{2y}$.

[2] Separation of Variables: Divide by e^{2y} (multiply by e^{-2y}) and multiply by dx : $e^{-2y} dy = e^{3x} dx$.

[3] Integration & Clearing Fractions:

$$\int e^{-2y} dy = \int e^{3x} dx \implies -\frac{1}{2} e^{-2y} = \frac{1}{3} e^{3x} + c_0$$

Multiply the entire equation by -6 : $3e^{-2y} = -2e^{3x} - 6c_0 \implies 3e^{-2y} + 2e^{3x} = c$.

Official Book Answer: $3e^{-2y} + 2e^{3x} = c$ (or $-e^{-2y}/2 = e^{3x}/3 + c_0$)

[VERIFIED · ZILL SM]

PROBLEM 2.2.9 · Separation with Logarithms & Parts

INTEGRATION BY PARTS

[QUESTION STATEMENT] Solve the given differential equation by separation of variables: $y \ln x \frac{dy}{dx} = \left(\frac{y+1}{x}\right)^2$.

[1] Separation: Multiply by x^2 , divide by $(y+1)^2$, and cross-multiply differentials:

$$x^2 \ln x dx = \frac{(y+1)^2}{y} dy \implies \left(y + 2 + \frac{1}{y}\right) dy = x^2 \ln x dx$$

[2] Integration by Parts on x : Let $u = \ln x$, $dv = x^2 dx \implies du = \frac{1}{x} dx$, $v = \frac{1}{3} x^3$:

$$\int x^2 \ln x dx = \frac{x^3}{3} \ln x - \int \frac{x^3}{3} \cdot \frac{1}{x} dx = \frac{1}{3} x^3 \ln |x| - \frac{1}{9} x^3$$

[3] Integration on y : $\int \left(y + 2 + \frac{1}{y}\right) dy = \frac{1}{2} y^2 + 2y + \ln |y|$. Equating gives:

$$\frac{1}{2} y^2 + 2y + \ln |y| = \frac{1}{3} x^3 \ln |x| - \frac{1}{9} x^3 + c$$

Official Book Answer: $\frac{1}{2} y^2 + 2y + \ln |y| = \frac{1}{3} x^3 \ln |x| - \frac{1}{9} x^3 + c$

[VERIFIED · ZILL SM]

PROBLEM 2.2.13 · Power Rule Separation

SUBSTITUTION

[QUESTION STATEMENT] Solve: $(e^y + 1)^2 e^{-y} dx + (e^x + 1)^3 e^{-x} dy = 0$.

[1] Separation: Multiply by $e^y e^x$ and rearrange: $\frac{e^y}{(e^y+1)^2} dy = -\frac{e^x}{(e^x+1)^3} dx$.

[2] Integration: Let $u = e^y + 1 \implies du = e^y dy$, and $w = e^x + 1 \implies dw = e^x dx$:

$$\int u^{-2} du = -\int w^{-3} dw \implies -u^{-1} = -\left(-\frac{1}{2} w^{-2}\right) + c \implies -(e^y + 1)^{-1} = \frac{1}{2} (e^x + 1)^{-2} + c$$

Official Book Answer: $-(e^y + 1)^{-1} = \frac{1}{2} (e^x + 1)^{-2} + c$ (or $\frac{1}{e^y+1} + \frac{1}{2(e^x+1)^2} = C$)

[VERIFIED · ZILL SM]

SECTION 2.2 · FACTORING, INITIAL-VALUE PROBLEMS & SINGULAR SOLUTIONS

Problems 19, 25, 27, 31

SEPARABLE

PROBLEM 2.2.19 · Factoring by Grouping

[QUESTION STATEMENT] Solve: $\frac{dy}{dx} = \frac{xy+3x-y-3}{xy-2x+4y-8}$.**[1] Factoring Numerator & Denominator:**Numerator: $x(y+3) - (y+3) = (x-1)(y+3)$. Denominator: $x(y-2) + 4(y-2) = (x+4)(y-2)$.

$$\frac{dy}{dx} = \frac{(x-1)(y+3)}{(x+4)(y-2)} \Rightarrow \frac{y-2}{y+3} dy = \frac{x-1}{x+4} dx$$

[2] Polynomial Division & Integration: Rewrite: $\frac{y-2}{y+3} = 1 - \frac{5}{y+3}$ and $\frac{x-1}{x+4} = 1 - \frac{5}{x+4}$.

$$\int \left(1 - \frac{5}{y+3}\right) dy = \int \left(1 - \frac{5}{x+4}\right) dx \Rightarrow y - 5 \ln|y+3| = x - 5 \ln|x+4| + c$$

[3] Exponentiation to Book Form: $5 \ln|x+4| - 5 \ln|y+3| = x - y + c \Rightarrow \ln \left| \frac{x+4}{y+3} \right|^5 = x - y + c \Rightarrow \left(\frac{x+4}{y+3} \right)^5 = c_1 e^{x-y}$.**Official Book Answer:** $\left(\frac{x+4}{y+3} \right)^5 = c_1 e^{x-y}$ (or $y - 5 \ln|y+3| = x - 5 \ln|x+4| + c$)

[VERIFIED · ZILL SM]

PROBLEM 2.2.25 · Explicit Initial-Value Problem

IVP

[QUESTION STATEMENT] Find an explicit solution of the initial-value problem: $x^2 \frac{dy}{dx} = y - xy$, $y(-1) = -1$.**[1] Factoring & Separation:** $x^2 \frac{dy}{dx} = y(1-x) \Rightarrow \frac{1}{y} dy = \frac{1-x}{x^2} dx = (x^{-2} - x^{-1}) dx$.**[2] General Solution:**

$$\ln|y| = -\frac{1}{x} - \ln|x| + c \Rightarrow \ln|y| + \ln|x| = -\frac{1}{x} + c \Rightarrow \ln|xy| = -\frac{1}{x} + c \Rightarrow xy = c_1 e^{-1/x}$$

[3] Initial Condition $y(-1) = -1$: Substitute $x = -1, y = -1$:

$$(-1)(-1) = 1 = c_1 e^{-1/(-1)} = c_1 e^1 \Rightarrow c_1 = e^{-1}$$

Thus $xy = e^{-1} e^{-1/x} = e^{-1-1/x}$. Explicit form: $y(x) = \frac{1}{x} e^{-1-1/x}$ defined on $(-\infty, 0)$.**Official Book Answer:** $xy = e^{-1-1/x}$ (explicit: $y = \frac{1}{x} e^{-(x+1)/x}$ on $(-\infty, 0)$)

[VERIFIED · ZILL SM]

PROBLEM 2.2.27 · Inverse Trig IVP & Addition Identity

TRIGONOMETRIC IVP

[QUESTION STATEMENT] Find an explicit solution of: $\sqrt{1-y^2} dx - \sqrt{1-x^2} dy = 0$, $y(0) = \frac{\sqrt{3}}{2}$.**[1] Separation & Integration:** $\frac{dx}{\sqrt{1-x^2}} - \frac{dy}{\sqrt{1-y^2}} = 0 \Rightarrow \arcsin x - \arcsin y = c$.**[2] Initial Condition:** $x = 0, y = \sqrt{3}/2 \Rightarrow \arcsin(0) - \arcsin(\sqrt{3}/2) = c \Rightarrow 0 - \pi/3 = c \Rightarrow c = -\pi/3$.

$$\arcsin y - \arcsin x = \frac{\pi}{3} \Rightarrow \arcsin y = \arcsin x + \frac{\pi}{3}$$

[3] Explicit Reduction via Sine Addition Formula:

$$y = \sin \left(\arcsin x + \frac{\pi}{3} \right) = \sin(\arcsin x) \cos \frac{\pi}{3} + \cos(\arcsin x) \sin \frac{\pi}{3} = x \left(\frac{1}{2} \right) + \sqrt{1-x^2} \left(\frac{\sqrt{3}}{2} \right) = \frac{x + \sqrt{3}\sqrt{1-x^2}}{2}$$

Official Book Answer: $y = \frac{x}{2} + \frac{\sqrt{3}}{2} \sqrt{1-x^2}$ (implicit: $\arcsin x - \arcsin y = -\pi/3$)

[VERIFIED · ZILL SM]

PROBLEM 2.2.31 · Singular Solutions Lost in Separation

SINGULAR SOLUTIONS

*[7th Edition Problem 31 / 6th Edition Problem 31: Lost Constant Solutions]***[QUESTION STATEMENT]** Find the singular / constant solutions lost during division for: (a) $\frac{dy}{dx} = x\sqrt{1-y^2}$, (b) $(e^x + e^{-x}) \frac{dy}{dx} = y^2$.
(7th Ed alternate: Solve $\frac{dy}{dx} = y^2, y(0) = 1$ using the definite integral formulation $\int_{x_0}^x$.)**[1] Singular Solution Analysis (6th Ed #31):**

• For $\frac{dy}{dx} = x\sqrt{1-y^2}$, separating variables requires dividing by $\sqrt{1-y^2}$. We must assume $1-y^2 \neq 0$. Setting $1-y^2 = 0$ gives $y = 1$ and $y = -1$. Checking the DE: $\frac{d}{dx}(\pm 1) = 0$ and $x\sqrt{1-(\pm 1)^2} = 0$. Both $y = 1$ and $y = -1$ are valid constant solutions that cannot be obtained from the general family for any finite c . They are **singular solutions**.

• For $(e^x + e^{-x}) \frac{dy}{dx} = y^2$, dividing by y^2 assumes $y \neq 0$. Setting $y = 0 \Rightarrow y' = 0$, which satisfies the DE $0 = 0$. Thus $y = 0$ is a **singular solution**.

[2] 7th Edition Integral Formulation: $\frac{dy}{y^2} = dx \Rightarrow -\frac{1}{y} + 1 = x \Rightarrow y = \frac{1}{1-x}$, defined on $(-\infty, 1)$.**Official Book Answer:** Singular solutions: For $\frac{dy}{dx} = x\sqrt{1-y^2}$, singular solutions are $y = -1$ and $y = 1$. For $(e^x + e^{-x})y' = y^2$, singular solution is $y = 0$.

[VERIFIED · ZILL SM]

SECTION 2.3 · LINEAR FIRST-ORDER DIFFERENTIAL EQUATIONS

Problems 4, 7, 9

LINEAR ODE

PROBLEM 2.3.4 · Constant Coefficients & Transient Term

[QUESTION STATEMENT] Find the general solution of the given differential equation. Give the largest interval over which the general solution is defined. Determine whether there are any transient terms:

$$\frac{dy}{dx} + 4y = \frac{4}{3}$$

[1] Standard Form & Integrating Factor: The equation is in standard form $y' + P(x)y = f(x)$ with $P(x) = 4$ and $f(x) = 4/3$. Both are continuous everywhere $(-\infty, \infty)$.

$$\mu(x) = e^{\int P(x) dx} = e^{\int 4 dx} = e^{4x}$$

[2] Exact Derivative & Integration: Multiply the ODE by $\mu(x) = e^{4x}$:

$$\frac{d}{dx} [e^{4x} y] = \frac{4}{3} e^{4x} \implies e^{4x} y = \int \frac{4}{3} e^{4x} dx = \frac{4}{3} \left(\frac{1}{4} e^{4x} \right) + c = \frac{1}{3} e^{4x} + c$$

[3] General Solution & Interval: Multiply by e^{-4x} :

$$y(x) = \frac{1}{3} + ce^{-4x}, \quad I = (-\infty, \infty)$$

Since $\lim_{x \rightarrow \infty} ce^{-4x} = 0$, the term ce^{-4x} is a **transient term** (the steady-state solution is $y_{\text{steady}} = 1/3$).

Official Book Answer: $y = \frac{1}{3} + ce^{-4x}$ for $-\infty < x < \infty$. Transient term is ce^{-4x} .

[VERIFIED · ZILL SM]

PROBLEM 2.3.7 · Singular Point at Origin

INTEGRATING FACTOR

[QUESTION STATEMENT] Find the general solution and largest interval of definition: $x \frac{dy}{dx} + y = \frac{1}{x}$.

[1] Standard Form: Divide by x (noting singular point at $x = 0$):

$$\frac{dy}{dx} + \frac{1}{x}y = \frac{1}{x^2}$$

Here $P(x) = 1/x$ and $f(x) = 1/x^2$, which are continuous on $(0, \infty)$ or $(-\infty, 0)$. We choose $I = (0, \infty)$.

[2] Integrating Factor: $\mu(x) = e^{\int (1/x) dx} = e^{\ln x} = x$.

[3] Integration & Solution:

$$\frac{d}{dx} [xy] = x \cdot \frac{1}{x^2} = \frac{1}{x} \implies xy = \int \frac{1}{x} dx = \ln x + c \implies y(x) = \frac{\ln x}{x} + \frac{c}{x}$$

Official Book Answer: $y = \frac{1}{x} \ln x + \frac{c}{x}$ for $0 < x < \infty$.

[VERIFIED · ZILL SM]

PROBLEM 2.3.9 · Negative Exponent Integrating Factor

TRIG PRODUCT

[QUESTION STATEMENT] Find the general solution and largest interval: $x \frac{dy}{dx} - y = x^2 \sin x$.

[1] Standard Form: Divide by x : $\frac{dy}{dx} - \frac{1}{x}y = x \sin x$. Continuous on $(0, \infty)$.

[2] Integrating Factor: $\mu(x) = e^{\int (-1/x) dx} = e^{-\ln x} = e^{\ln(x^{-1})} = \frac{1}{x}$.

[3] Integration:

$$\frac{d}{dx} \left[\frac{1}{x} y \right] = \frac{1}{x} (x \sin x) = \sin x \implies \frac{1}{x} y = -\cos x + c \implies y(x) = cx - x \cos x$$

Official Book Answer: $y = cx - x \cos x$ for $0 < x < \infty$.

[VERIFIED · ZILL SM]

SECTION 2.3 · ENGINEERING APPLICATIONS & DISCONTINUOUS DRIVING FUNCTIONS

Problems 23, 27, 31

LINEAR ODE

PROBLEM 2.3.23 · Combined Exponential & Power Factor

[QUESTION STATEMENT] Find the general solution and transient terms for: $x \frac{dy}{dx} + (3x + 1)y = e^{-3x}$.**[1] Standard Form:** $\frac{dy}{dx} + \left(3 + \frac{1}{x}\right)y = \frac{e^{-3x}}{x}$. Continuous on $(0, \infty)$.**[2] Integrating Factor:** $\mu(x) = e^{\int (3+1/x) dx} = e^{3x+\ln x} = e^{3x} \cdot x = xe^{3x}$.**[3] Integration & Solution:**

$$\frac{d}{dx} [xe^{3x}y] = xe^{3x} \cdot \frac{e^{-3x}}{x} = 1 \implies xe^{3x}y = x + c \implies y(x) = e^{-3x} + \frac{ce^{-3x}}{x}$$

Largest interval is $I = (0, \infty)$. Since $\lim_{x \rightarrow \infty} y(x) = 0$, the entire expression (and specifically ce^{-3x}/x) represents transient behavior.**Official Book Answer:** $y = e^{-3x} + \frac{ce^{-3x}}{x}$ for $0 < x < \infty$. Transient term is $\frac{ce^{-3x}}{x}$.

[VERIFIED · ZILL SM]

PROBLEM 2.3.27 · Series RL Circuit Response

RL CIRCUIT

[QUESTION STATEMENT] Solve the initial-value problem for a series RL circuit: $L \frac{di}{dt} + Ri = E$, $i(0) = i_0$, where L, R, E , and i_0 are constants.**[1] Standard Form:** $\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$. Integrating factor: $\mu(t) = e^{\int (R/L) dt} = e^{Rt/L}$.**[2] General Solution:**

$$\frac{d}{dt} [ie^{Rt/L}] = \frac{E}{L}e^{Rt/L} \implies ie^{Rt/L} = \frac{E}{L} \left(\frac{L}{R} e^{Rt/L} \right) + c = \frac{E}{R} e^{Rt/L} + c \implies i(t) = \frac{E}{R} + ce^{-Rt/L}$$

[3] Initial Condition $i(0) = i_0$:

$$i_0 = \frac{E}{R} + c \implies c = i_0 - \frac{E}{R} \implies i(t) = \frac{E}{R} + \left(i_0 - \frac{E}{R}\right)e^{-Rt/L}$$

Physical interpretation: $\frac{E}{R}$ is the steady-state current; $\left(i_0 - \frac{E}{R}\right)e^{-Rt/L}$ is the transient current decaying with time constant $\tau = L/R$.**Official Book Answer:** $i(t) = \frac{E}{R} + \left(i_0 - \frac{E}{R}\right)e^{-Rt/L}$ for $-\infty < t < \infty$.

[VERIFIED · ZILL SM]

PROBLEM 2.3.31 · Piecewise Discontinuous Input

PIECEWISE IVP

[QUESTION STATEMENT] Solve the initial-value problem: $\frac{dy}{dx} + 2y = f(x)$, $y(0) = 0$, where $f(x) = \begin{cases} 1, & 0 \leq x \leq 3 \\ 0, & x > 3 \end{cases}$. Give a continuous solution.**[1] Interval** $0 \leq x \leq 3$: $\frac{dy}{dx} + 2y = 1$. Integrating factor is e^{2x} :

$$\frac{d}{dx} [e^{2x}y] = e^{2x} \implies e^{2x}y = \frac{1}{2}e^{2x} + c_1 \implies y(x) = \frac{1}{2} + c_1e^{-2x}$$

Applying $y(0) = 0 \implies 0 = \frac{1}{2} + c_1 \implies c_1 = -\frac{1}{2} \implies y(x) = \frac{1}{2}(1 - e^{-2x})$.**[2] Interval** $x > 3$: $\frac{dy}{dx} + 2y = 0 \implies y(x) = c_2e^{-2x}$.**[3] Enforcing Continuity at** $x = 3$:

$$\lim_{x \rightarrow 3^-} y(x) = \frac{1}{2}(1 - e^{-6}) \quad \text{must equal} \quad \lim_{x \rightarrow 3^+} y(x) = c_2e^{-6}$$

$$c_2e^{-6} = \frac{1}{2}(1 - e^{-6}) \implies c_2 = \frac{1}{2}(1 - e^{-6})e^6 = \frac{1}{2}(e^6 - 1)$$

Official Book Answer: $y(x) = \begin{cases} \frac{1}{2}(1 - e^{-2x}), & 0 \leq x \leq 3 \\ \frac{1}{2}(e^6 - 1)e^{-2x}, & x > 3 \end{cases}$

[VERIFIED · ZILL SM]

SECTION 2.4 · EXACT DIFFERENTIAL EQUATIONS

Problems 3, 5, 9

PROBLEM 2.4.3 · Basic Test for Exactness

EXACT DE

[QUESTION STATEMENT] Determine whether the given differential equation is exact. If it is exact, solve it:

$$(5x + 4y) dx + (4x - 8y^3) dy = 0$$

[1] Test for Exactness: Identify $M(x, y) = 5x + 4y$ and $N(x, y) = 4x - 8y^3$.

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(5x + 4y) = 4, \quad \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(4x - 8y^3) = 4$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 4$ everywhere in $\mathbb{R}^2$, the equation is **exact**.**[2] Find Potential Function $f(x, y)$:**

$$f(x, y) = \int M(x, y) dx = \int (5x + 4y) dx = \frac{5}{2}x^2 + 4xy + g(y)$$

[3] Differentiate with respect to y and match N :

$$\frac{\partial f}{\partial y} = 4x + g'(y) = N(x, y) = 4x - 8y^3 \implies g'(y) = -8y^3 \implies g(y) = -2y^4$$

Therefore, $f(x, y) = \frac{5}{2}x^2 + 4xy - 2y^4$. Setting $f(x, y) = c$ gives the general implicit solution.**Official Book Answer:** $\frac{5}{2}x^2 + 4xy - 2y^4 = c$

[VERIFIED · ZILL SM]

PROBLEM 2.4.5 · Mixed Polynomial Exact Equation

EXACT DE

[QUESTION STATEMENT] Determine whether exact and solve: $(2xy^2 - 3) dx + (2x^2y + 4) dy = 0$.**[1] Test for Exactness:** $M = 2xy^2 - 3$, $N = 2x^2y + 4$.

$$\frac{\partial M}{\partial y} = 4xy, \quad \frac{\partial N}{\partial x} = 4xy \implies \text{Exact!}$$

[2] Potential Function:

$$f(x, y) = \int M dx = \int (2xy^2 - 3) dx = x^2y^2 - 3x + g(y)$$

$$\frac{\partial f}{\partial y} = 2x^2y + g'(y) = N = 2x^2y + 4 \implies g'(y) = 4 \implies g(y) = 4y$$

General solution: $x^2y^2 - 3x + 4y = c$.**Official Book Answer:** $x^2y^2 - 3x + 4y = c$

[VERIFIED · ZILL SM]

PROBLEM 2.4.9 · Trigonometric Exact Equation

TRIG EXACT

[QUESTION STATEMENT] Determine whether exact and solve: $(x - y^3 + y^2 \sin x) dx = (3xy^2 + 2y \cos x) dy$.**[1] Standard Differential Form:**

$$(y^3 - y^2 \sin x - x) dx + (3xy^2 + 2y \cos x) dy = 0$$

Identify $M = y^3 - y^2 \sin x - x$ and $N = 3xy^2 + 2y \cos x$.

$$\frac{\partial M}{\partial y} = 3y^2 - 2y \sin x, \quad \frac{\partial N}{\partial x} = 3y^2 - 2y \sin x \implies \text{Exact!}$$

[2] Integrate N with respect to y :

$$f(x, y) = \int (3xy^2 + 2y \cos x) dy = xy^3 + y^2 \cos x + h(x)$$

$$\frac{\partial f}{\partial x} = y^3 - y^2 \sin x + h'(x) = M = y^3 - y^2 \sin x - x \implies h'(x) = -x \implies h(x) = -\frac{1}{2}x^2$$

Official Book Answer: $xy^3 + y^2 \cos x - \frac{1}{2}x^2 = c$

[VERIFIED · ZILL SM]

SECTION 2.4 · INVERSE TANGENT, PARAMETER DETERMINATION & GIVEN FACTORS

Problems 15, 27, 29

PROBLEM 2.4.15 · Inverse Trigonometric Potential

EXACT DE

[QUESTION STATEMENT] Determine whether exact and solve: $(x^2y^3 - \frac{1}{1+9x^2}) dx + x^3y^2 dy = 0$.

[1] Exactness Test: $M = x^2y^3 - \frac{1}{1+9x^2}$, $N = x^3y^2$.

$$\frac{\partial M}{\partial y} = 3x^2y^2, \quad \frac{\partial N}{\partial x} = 3x^2y^2 \implies \text{Exact!}$$

[2] Integration on N :

$$\begin{aligned} f(x, y) &= \int N dy = \int x^3y^2 dy = \frac{1}{3}x^3y^3 + h(x) \\ \frac{\partial f}{\partial x} &= x^2y^3 + h'(x) = M = x^2y^3 - \frac{1}{1+9x^2} \implies h'(x) = -\frac{1}{1+(3x)^2} \\ h(x) &= -\int \frac{1}{1+(3x)^2} dx = -\frac{1}{3} \arctan(3x) \end{aligned}$$

Thus $f(x, y) = \frac{1}{3}x^3y^3 - \frac{1}{3} \arctan(3x) = c_0 \implies x^3y^3 - \arctan(3x) = c$.

Official Book Answer: $x^3y^3 - \arctan(3x) = c$

[VERIFIED · ZILL SM]

PROBLEM 2.4.27 · Parameter Determination for Exactness

PARAMETER K

[QUESTION STATEMENT] Find the value of k so that the given differential equation is exact:

$$(y^3 + kxy^4 - 2x) dx + (3xy^2 + 20x^2y^3) dy = 0$$

[1] Partial Derivatives:

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y}(y^3 + kxy^4 - 2x) = 3y^2 + 4kxy^3 \\ \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x}(3xy^2 + 20x^2y^3) = 3y^2 + 40xy^3 \end{aligned}$$

[2] Equate Partial: For exactness, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ for all (x, y) :

$$3y^2 + 4kxy^3 = 3y^2 + 40xy^3 \implies 4kxy^3 = 40xy^3 \implies 4k = 40 \implies k = 10$$

Concordia Exam Tip: This is a favorite midterm question. Always compute both partials fully and equate matching powers of x and y .

Official Book Answer: $k = 10$

[VERIFIED · ZILL SM]

PROBLEM 2.4.29 · Verification of Given Integrating Factor

INTEGRATING FACTOR

[QUESTION STATEMENT] Verify that $(-xy \sin x + 2y \cos x) dx + 2x \cos x dy = 0$ is not exact. Multiply by the given factor $\mu(x, y) = xy$, verify that the new equation is exact, and solve.

[1] Check Original Equation: $M_y = -x \sin x + 2 \cos x$, $N_x = 2 \cos x - 2x \sin x$. Since $M_y \neq N_x$, it is **not exact**.

[2] Multiply by $\mu(x, y) = xy$:

$$(-x^2y^2 \sin x + 2xy^2 \cos x) dx + (2x^2y \cos x) dy = 0$$

New $M_y = -2x^2y \sin x + 4xy \cos x$. New $N_x = \frac{\partial}{\partial x}(2x^2y \cos x) = 4xy \cos x - 2x^2y \sin x$. They match identically! Exact!

[3] Potential Function:

$$\begin{aligned} f(x, y) &= \int (2x^2y \cos x) dy = x^2y^2 \cos x + h(x) \\ \frac{\partial f}{\partial x} &= 2xy^2 \cos x - x^2y^2 \sin x + h'(x) = M \implies h'(x) = 0 \implies h(x) = 0 \end{aligned}$$

Official Book Answer: $x^2y^2 \cos x = c$

[VERIFIED · ZILL SM]

SECTION 2.4 · SPECIAL INTEGRATING FACTORS $M(X)$ & $M(Y)$, AND INITIAL-VALUE PROBLEMS**PROBLEM 2.4.33 · Integrating Factor Depending on x Alone**INTEGRATING FACTOR $M(X)$ **[QUESTION STATEMENT]** Solve the given differential equation by finding an appropriate integrating factor:

$$(2y^2 + 3x) dx + 2xy dy = 0$$

[1] Check Exactness: $M = 2y^2 + 3x$, $N = 2xy$. $\frac{\partial M}{\partial y} = 4y$, $\frac{\partial N}{\partial x} = 2y \implies$ Not exact.**[2] Formula for $\mu(x)$:**

$$\frac{M_y - N_x}{N} = \frac{4y - 2y}{2xy} = \frac{2y}{2xy} = \frac{1}{x} \quad (\text{function of } x \text{ alone!})$$

$$\mu(x) = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

[3] Multiply by $\mu(x) = x$ & Solve:

$$(2xy^2 + 3x^2) dx + 2x^2y dy = 0$$

$$f(x, y) = \int 2x^2y dy = x^2y^2 + h(x)$$

$$\frac{\partial f}{\partial x} = 2xy^2 + h'(x) = 2xy^2 + 3x^2 \implies h'(x) = 3x^2 \implies h(x) = x^3$$

Official Book Answer: $x^2y^2 + x^3 = c$

[VERIFIED · ZILL SM]

PROBLEM 2.4.35 · Integrating Factor Depending on y AloneINTEGRATING FACTOR $M(Y)$ **[QUESTION STATEMENT]** Solve: $6xy dx + (4y + 9x^2) dy = 0$.**[1] Check Exactness:** $M = 6xy$, $N = 4y + 9x^2$. $M_y = 6x$, $N_x = 18x \implies$ Not exact.**[2] Formula for $\mu(y)$:**

$$\frac{N_x - M_y}{M} = \frac{18x - 6x}{6xy} = \frac{12x}{6xy} = \frac{2}{y} \quad (\text{function of } y \text{ alone!})$$

$$\mu(y) = e^{\int \frac{2}{y} dy} = e^{2 \ln y} = y^2$$

[3] Multiply by $\mu(y) = y^2$ & Solve:

$$6xy^3 dx + (4y^3 + 9x^2y^2) dy = 0$$

$$f(x, y) = \int 6xy^3 dx = 3x^2y^3 + g(y)$$

$$\frac{\partial f}{\partial y} = 9x^2y^2 + g'(y) = 4y^3 + 9x^2y^2 \implies g'(y) = 4y^3 \implies g(y) = y^4$$

Official Book Answer: $3x^2y^3 + y^4 = c$

[VERIFIED · ZILL SM]

PROBLEM 2.4.39 · Exact Initial-Value Problem

IVP

[QUESTION STATEMENT] Solve the initial-value problem: $x dx + (x^2y + 4y) dy = 0$, $y(4) = 0$.**[1] Separation of Variables vs. Exact Integrating Factor:** Notice that $x dx + y(x^2 + 4) dy = 0$. Dividing by $x^2 + 4$ yields separable form:

$$\frac{x}{x^2 + 4} dx + y dy = 0$$

(Equivalently, the integrating factor for the original DE is $\mu(x) = \frac{1}{x^2 + 4}$).**[2] Integration:**

$$\int \frac{x}{x^2 + 4} dx + \int y dy = 0 \implies \frac{1}{2} \ln(x^2 + 4) + \frac{1}{2} y^2 = c_0 \implies \ln(x^2 + 4) + y^2 = c$$

[3] Apply Initial Condition $y(4) = 0$:

$$\ln(4^2 + 4) + 0^2 = c \implies \ln(20) = c \implies \ln(x^2 + 4) + y^2 = \ln 20$$

Exponentiating: $x^2 + 4 = 20e^{-y^2}$, or $y^2 = \ln\left(\frac{20}{x^2 + 4}\right)$.**Official Book Answer:** $\ln(x^2 + 4) + y^2 = \ln 20$ (or $x^2 + 4 = 20e^{-y^2}$)

[VERIFIED · ZILL SM]

SECTION 2.5 · HOMOGENEOUS DIFFERENTIAL EQUATIONS ($Y = UX$)

Problems 5, 7, 9, 13

PROBLEM 2.5.5 · Homogeneous Degree 2

HOMOGENEOUS DE

[QUESTION STATEMENT] Solve: $(y^2 + yx) dx - x^2 dy = 0$.[1] Degree Check & Substitution: $M(tx, ty) = t^2 M(x, y)$ and $N(tx, ty) = t^2 N(x, y)$, so both coefficients are homogeneous of degree 2. Let $y = ux \implies dy = u dx + x du$.

[2] Algebraic Substitution:

$$(u^2 x^2 + ux^2) dx - x^2(u dx + x du) = 0 \implies (u^2 + u - u)x^2 dx - x^3 du = 0 \implies u^2 x^2 dx - x^3 du = 0$$

[3] Separation & Resubstitution:

$$\frac{dx}{x} - \frac{du}{u^2} = 0 \implies \ln|x| + \frac{1}{u} = c \implies \ln|x| + \frac{x}{y} = c \implies y \ln|x| + x = cy$$

Official Book Answer: $y \ln|x| + x = cy$

[VERIFIED · ZILL SM]

PROBLEM 2.5.7 · Homogeneous Degree 1 with Arctan

HOMOGENEOUS DE

[QUESTION STATEMENT] Solve: $(y - x) dx - (y + x) dy = 0$.[1] Substitution $y = ux, dy = u dx + x du$:

$$(u x - x) dx - (u x + x)(u dx + x du) = 0 \implies x(u - 1 - u^2 - u) dx - x^2(u + 1) du = 0$$

$$-(u^2 + 1) dx - x(u + 1) du = 0 \implies \frac{dx}{x} + \frac{u + 1}{u^2 + 1} du = 0$$

[2] Split & Integrate:

$$\int \frac{dx}{x} + \int \frac{u}{u^2 + 1} du + \int \frac{1}{u^2 + 1} du = 0 \implies \ln|x| + \frac{1}{2} \ln(u^2 + 1) + \arctan u = c$$

[3] Combine Logarithms:

$$\ln[x^2(u^2 + 1)] + 2 \arctan u = c_1 \implies \ln\left[x^2\left(\frac{y^2}{x^2} + 1\right)\right] + 2 \arctan\left(\frac{y}{x}\right) = c_1 \implies \ln(x^2 + y^2) + 2 \arctan\left(\frac{y}{x}\right) = c_1$$

Official Book Answer: $\ln(x^2 + y^2) + 2 \arctan(y/x) = c_1$

[VERIFIED · ZILL SM]

PROBLEM 2.5.9 · Radical Homogeneous Equation

RADICAL FORM

[QUESTION STATEMENT] Solve: $-y dx + (x + \sqrt{xy}) dy = 0$.[1] Substitution $y = ux, dy = u dx + x du$:

$$-ux dx + (x + x\sqrt{u})(u dx + x du) = 0 \implies (-u + u + u^{3/2})x dx + x^2(1 + \sqrt{u}) du = 0$$

$$u^{3/2} dx + x(1 + u^{1/2}) du = 0 \implies \frac{dx}{x} + \frac{1 + u^{1/2}}{u^{3/2}} du = 0 \implies \frac{dx}{x} + \left(u^{-3/2} + \frac{1}{u}\right) du = 0$$

[2] Integration & Resubstitution:

$$\ln|x| - 2u^{-1/2} + \ln|u| = c \implies \ln|x \cdot u| - 2u^{-1/2} = c \implies \ln|y| - c = 2\sqrt{\frac{x}{y}} \implies y(\ln|y| - c)^2 = 4x$$

Official Book Answer: $y(\ln|y| - c)^2 = 4x$

[VERIFIED · ZILL SM]

PROBLEM 2.5.13 · Exponential Homogeneous IVP

IVP

[QUESTION STATEMENT] Solve the initial-value problem: $(x + ye^{y/x}) dx - xe^{y/x} dy = 0, \quad y(1) = 0$.[1] Substitution $y = ux, dy = u dx + x du$:

$$(x + uxe^u) dx - xe^u(u dx + x du) = 0 \implies x dx - x^2 e^u du = 0 \implies \frac{dx}{x} - e^u du = 0$$

[2] General Solution: $\ln|x| - e^u = c \implies \ln|x| - e^{y/x} = c$.[3] Initial Condition $y(1) = 0$: $\ln(1) - e^0 = c \implies 0 - 1 = c \implies c = -1$.

$$\ln|x| - e^{y/x} = -1 \implies \ln|x| = e^{y/x} - 1 \quad (\text{or } y = x \ln(1 + \ln|x|))$$

Official Book Answer: $\ln|x| = e^{y/x} - 1$

[VERIFIED · ZILL SM]

SECTION 2.5 · BERNOULLI DIFFERENTIAL EQUATIONS ($U = Y^{1-N}$)**BERNOULLI DE****PROBLEM 2.5.17 · Bernoulli $n = 4$** **[QUESTION STATEMENT]** Solve the given Bernoulli differential equation: $y' + y = xy^4$.**[1] Standard Bernoulli Form:** Here $P(x) = 1$, $f(x) = x$, and the nonlinear power is $n = 4$. Divide by y^4 : $y^{-4}y' + y^{-3} = x$.**[2] Linearizing Substitution:** Let $w = y^{1-n} = y^{1-4} = y^{-3}$.

$$\frac{dw}{dx} = -3y^{-4} \frac{dy}{dx} \implies y^{-4} \frac{dy}{dx} = -\frac{1}{3} \frac{dw}{dx}$$

Substitute into DE: $-\frac{1}{3} \frac{dw}{dx} + w = x \implies \frac{dw}{dx} - 3w = -3x$.**[3] Linear Integrating Factor & Parts:** $\mu(x) = e^{\int -3 dx} = e^{-3x}$.

$$\frac{d}{dx} [we^{-3x}] = -3xe^{-3x} \implies we^{-3x} = \int -3xe^{-3x} dx = xe^{-3x} + \frac{1}{3}e^{-3x} + c$$

Divide by e^{-3x} : $w = x + \frac{1}{3} + ce^{3x}$. Since $w = y^{-3}$:

$$y^{-3} = x + \frac{1}{3} + ce^{3x} \quad \left(\text{or } y(x) = \frac{1}{\sqrt[3]{x + 1/3 + ce^{3x}}} \right)$$

Official Book Answer: $y^{-3} = x + \frac{1}{3} + ce^{3x}$ **[VERIFIED · ZILL SM]****PROBLEM 2.5.19 · Bernoulli $n = 2$ with Independent Variable t** **BERNOULLI DE****[QUESTION STATEMENT]** Solve: $t \frac{dy}{dt} - y = -\frac{y^2}{t}$.**[1] Standard Form:** Divide by t : $\frac{dy}{dt} - \frac{1}{t}y = -\frac{1}{t^2}y^2$. Here $n = 2$. Divide by y^2 : $y^{-2} \frac{dy}{dt} - \frac{1}{t}y^{-1} = -\frac{1}{t^2}$.**[2] Linearizing Substitution** $w = y^{-1}$: $\frac{dw}{dt} = -y^{-2} \frac{dy}{dt} \implies y^{-2} \frac{dy}{dt} = -\frac{dw}{dt}$.

$$-\frac{dw}{dt} - \frac{1}{t}w = -\frac{1}{t^2} \implies \frac{dw}{dt} + \frac{1}{t}w = \frac{1}{t^2}$$

[3] Solve Linear Equation: $\mu(t) = e^{\int (1/t) dt} = t$.

$$\frac{d}{dt} [tw] = t \cdot \frac{1}{t^2} = \frac{1}{t} \implies tw = \ln t + c \implies w = \frac{\ln t}{t} + \frac{c}{t}$$

Substitute back $w = y^{-1}$: $y^{-1} = \frac{\ln t + c}{t} \implies \frac{t}{y} = \ln t + c \implies e^{t/y} = c_1 t$.**Official Book Answer:** $y^{-1} = \frac{1}{t} \ln t + \frac{c}{t}$ (or $\frac{t}{y} = \ln t + c$, or $e^{t/y} = c_1 t$)**[VERIFIED · ZILL SM]****PROBLEM 2.5.21 · Bernoulli Initial-Value Problem****BERNOULLI IVP****[QUESTION STATEMENT]** Solve the initial-value problem: $x \frac{dy}{dx} - 2y = \frac{3}{x}y^4$, $y(1) = \frac{1}{2}$.**[1] Standard Form:** $\frac{dy}{dx} - \frac{2}{x}y = \frac{3}{x^2}y^4$. Divide by y^4 : $y^{-4} \frac{dy}{dx} - \frac{2}{x}y^{-3} = \frac{3}{x^2}$.**[2] Substitution** $w = y^{-3}$: $\frac{dw}{dx} = -3y^{-4}y' \implies -\frac{1}{3} \frac{dw}{dx} - \frac{2}{x}w = \frac{3}{x^2} \implies \frac{dw}{dx} + \frac{6}{x}w = -\frac{9}{x^2}$.

$$\mu(x) = e^{\int (6/x) dx} = x^6 \implies \frac{d}{dx} [x^6 w] = -9x^4 \implies x^6 w = -\frac{9}{5}x^5 + c \implies w = -\frac{9}{5}x^{-1} + cx^{-6}$$

[3] Initial Condition $y(1) = 1/2$: Since $w = y^{-3}$, $w(1) = (1/2)^{-3} = 8$.

$$8 = -\frac{9}{5}(1) + c(1) \implies c = 8 + \frac{9}{5} = \frac{49}{5}$$

$$y^{-3} = -\frac{9}{5}x^{-1} + \frac{49}{5}x^{-6} \implies y(x) = \left(-\frac{9}{5x} + \frac{49}{5x^6} \right)^{-1/3} = \left(\frac{49 - 9x^5}{5x^6} \right)^{-1/3}$$

Official Book Answer: $y^{-3} = -\frac{9}{5}x^{-1} + \frac{49}{5}x^{-6}$ (or $y = \left(\frac{5x^6}{49 - 9x^5} \right)^{1/3}$)**[VERIFIED · ZILL SM]**

SECTION 2.5 · REDUCTION TO SEPARATION OF VARIABLES ($U = AX + BY + C$)**LINEAR SUBSTITUTION****PROBLEM 2.5.23 · Linear Substitution $u = x + y + 1$**

[QUESTION STATEMENT] Solve the differential equation: $\frac{dy}{dx} = (x + y + 1)^2$.

[1] Substitution: Let $u = x + y + 1$. Differentiating with respect to x gives:

$$\frac{du}{dx} = 1 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} - 1$$

[2] Separable DE in u :

$$\frac{du}{dx} - 1 = u^2 \implies \frac{du}{dx} = u^2 + 1 \implies \frac{du}{u^2 + 1} = dx$$

[3] Integration & Back-Substitution:

$$\int \frac{du}{u^2 + 1} = \int dx \implies \arctan u = x + c \implies u = \tan(x + c)$$

$$x + y + 1 = \tan(x + c) \implies y(x) = \tan(x + c) - x - 1$$

Official Book Answer: $y = \tan(x + c) - x - 1$

[VERIFIED · ZILL SM]

PROBLEM 2.5.25 · Linear Substitution with Trigonometry**LINEAR SUBSTITUTION**

[QUESTION STATEMENT] Solve: $\frac{dy}{dx} = \tan^2(x + y)$.

[1] Substitution: Let $u = x + y \implies \frac{du}{dx} = 1 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} - 1$.

[2] Separation & Identity:

$$\frac{du}{dx} - 1 = \tan^2 u \implies \frac{du}{dx} = 1 + \tan^2 u = \sec^2 u \implies \frac{du}{\sec^2 u} = dx \implies \cos^2 u du = dx$$

[3] Half-Angle Integration:

$$\int \frac{1 + \cos 2u}{2} du = \int dx \implies \frac{1}{2}u + \frac{1}{4}\sin 2u = x + c \implies 2u + \sin 2u = 4x + c_1$$

Substitute $u = x + y$:

$$2(x + y) + \sin(2(x + y)) = 4x + c_1 \implies 2y + \sin(2(x + y)) = 2x + c_1$$

Official Book Answer: $2y + \sin(2(x + y)) = 2x + c_1$

[VERIFIED · ZILL SM]

PROBLEM 2.5.27 · Radical Linear Substitution**SQUARE ROOT**

[QUESTION STATEMENT] Solve: $\frac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$.

[1] Substitution: Let $u = y - 2x + 3 \implies \frac{du}{dx} = \frac{dy}{dx} - 2 \implies \frac{dy}{dx} = \frac{du}{dx} + 2$.

[2] Separation:

$$\frac{du}{dx} + 2 = 2 + \sqrt{u} \implies \frac{du}{dx} = \sqrt{u} \implies u^{-1/2} du = dx$$

[3] Integration & Resubstitution:

$$\int u^{-1/2} du = \int dx \implies 2\sqrt{u} = x + c \implies 2\sqrt{y - 2x + 3} = x + c$$

Squaring both sides (if explicit form desired): $y - 2x + 3 = \frac{1}{4}(x + c)^2 \implies y = 2x - 3 + \frac{1}{4}(x + c)^2$.

Official Book Answer: $2\sqrt{y - 2x + 3} = x + c$ (or $y = 2x - 3 + \frac{1}{4}(x + c)^2$)

[VERIFIED · ZILL SM]

SECTION 2.7 · LINEAR MODELS: POPULATION GROWTH, DECAY & THERMAL TRANSIENTS

Problems 3, 5, 9, 15

PROBLEM 2.7.3 · Malthusian Population Growth

EXPONENTIAL GROWTH

[QUESTION STATEMENT] A culture initially has $P_0 = 500$ bacteria. At $t = 10$ hours, the population is $P(10) = 575$. If the rate of growth is proportional to the population present at time t , determine the population at $t = 30$ hours.

[1] Model & General Solution: $\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt} = 500e^{kt}$.

[2] Determine Growth Constant k : At $t = 10$:

$$575 = 500e^{10k} \implies e^{10k} = \frac{575}{500} = 1.15 \implies k = \frac{1}{10} \ln(1.15) \approx 0.013976 \text{ hr}^{-1}$$

[3] Evaluate at $t = 30$ hours:

$$P(30) = 500e^{30k} = 500(e^{10k})^3 = 500(1.15)^3 = 500(1.520875) = 760.44 \approx 760 \text{ bacteria}$$

Official Book Answer: $P(30) \approx 760$ bacteria (exact: $500(1.15)^3$)

[VERIFIED · ZILL SM]

PROBLEM 2.7.5 · Light Extinction in Ocean Water

BEER-LAMBERT LAW

[QUESTION STATEMENT] The intensity of light I at depth x in ocean water satisfies $\frac{dI}{dx} = kI$. If I_0 is the surface intensity and light intensity drops to 25% of I_0 at depth $x = 3$ ft, find the intensity at $x = 15$ ft.

[1] Model & Calibration: $I(x) = I_0 e^{kx}$. At $x = 3$:

$$I(3) = 0.25I_0 \implies I_0 e^{3k} = 0.25I_0 \implies e^{3k} = 0.25 = \frac{1}{4} \implies k = \frac{1}{3} \ln(0.25)$$

[2] Compute Intensity at $x = 15$ ft:

$$I(15) = I_0 e^{15k} = I_0 (e^{3k})^5 = I_0 (0.25)^5 = I_0 \left(\frac{1}{4}\right)^5 = \frac{I_0}{1024} \approx 0.00097656I_0 \approx 0.00098I_0$$

Official Book Answer: $I(15) = (0.25)^5 I_0 \approx 0.00098I_0$ (less than 0.1% of surface intensity)

[VERIFIED · ZILL SM]

PROBLEM 2.7.9 · Radioactive Tracer Half-Life

RADIOACTIVE DECAY

[QUESTION STATEMENT] A radioactive tracer has 97% of its original mass remaining after 6 hours. Find the half-life of the substance, and determine when 50% has decayed.

[1] Decay Law: $N(t) = N_0 e^{kt}$. Using $N(6) = 0.97N_0$:

$$e^{6k} = 0.97 \implies k = \frac{1}{6} \ln(0.97) \approx -0.005076 \text{ hr}^{-1}$$

[2] Half-Life ($N(t_{1/2}) = 0.5N_0$):

$$0.5 = e^{kt} \implies kt = \ln(0.5) = -\ln 2 \implies t = \frac{-\ln 2}{(1/6)\ln(0.97)} \approx \frac{-0.693147}{-0.005076} \approx 136.5 \text{ hours}$$

Official Book Answer: $t = \frac{\ln(1/2)}{\frac{1}{6}\ln(0.97)} \approx 136.5$ hours (approx. 5.7 days)

[VERIFIED · ZILL SM]

PROBLEM 2.7.15 · Newton's Law of Heating

HEAT TRANSFER

[QUESTION STATEMENT] A thermometer reading 20°C is placed in boiling water (100°C). After 1 second, the thermometer reads 22°C . How long will it take to reach 90°C ? How long to reach 98°C ?

[1] Differential Equation & Integration:

$$\frac{dT}{dt} = k(T - 100) \implies T(t) = 100 + ce^{kt}$$

At $t = 0$, $T(0) = 20 \implies 20 = 100 + c \implies c = -80 \implies T(t) = 100 - 80e^{kt}$.

[2] Determine Rate Constant k : $T(1) = 22 \implies 22 = 100 - 80e^k \implies 80e^k = 78 \implies e^k = \frac{78}{80} = \frac{39}{40} \implies k = \ln(39/40) \approx -0.02532 \text{ s}^{-1}$.

[3] Target Temperatures:

- For $T = 90^\circ$: $90 = 100 - 80e^{kt} \implies 80e^{kt} = 10 \implies e^{kt} = \frac{1}{8} \implies t = \frac{\ln(1/8)}{\ln(39/40)} = \frac{-\ln 8}{-0.02532} \approx 82.1 \text{ s}$.
- For $T = 98^\circ$: $98 = 100 - 80e^{kt} \implies 80e^{kt} = 2 \implies e^{kt} = \frac{1}{40} \implies t = \frac{\ln(1/40)}{\ln(39/40)} = \frac{-\ln 40}{-0.02532} \approx 145.7 \text{ s}$.

Official Book Answer: $t(90^\circ\text{C}) \approx 82.1$ seconds; $t(98^\circ\text{C}) \approx 145.7$ seconds.

[VERIFIED · ZILL SM]

SECTION 2.7 · FORENSICS, VARIABLE-VOLUME TANKS & RL CIRCUITS

Problems 17, 23, 25

FORENSICS MODEL

PROBLEM 2.7.17 · Forensic Time of Death Estimation

[QUESTION STATEMENT] A murder victim was discovered in a room whose temperature is kept at a constant 65°F . At the time of discovery ($t = 0$), the core body temperature was 82°F . Two hours later ($t = 2$), the temperature was 75°F . Assuming normal body temperature at death was 98.6°F , estimate how long before discovery the murder occurred.

[1] Model Setup: By Newton's law of cooling: $\frac{dT}{dt} = k(T - 65) \implies T(t) = 65 + ce^{kt}$. At discovery $t = 0$: $T(0) = 82 \implies 82 = 65 + c \implies c = 17 \implies T(t) = 65 + 17e^{kt}$.

[2] Determine Rate Constant k : At $t = 2$ hours:

$$75 = 65 + 17e^{2k} \implies 17e^{2k} = 10 \implies e^{2k} = \frac{10}{17} \implies k = \frac{1}{2} \ln\left(\frac{10}{17}\right) \approx -0.2653 \text{ hr}^{-1}$$

Thus: $T(t) = 65 + 17e^{-0.2653t}$.

[3] Time of Death ($T(t_0) = 98.6^\circ\text{F}$):

$$98.6 = 65 + 17e^{-0.2653t_0} \implies 17e^{-0.2653t_0} = 33.6 \implies e^{-0.2653t_0} = \frac{33.6}{17} \approx 1.97647$$

$$-0.2653t_0 = \ln(1.97647) \approx 0.6813 \implies t_0 = -\frac{0.6813}{0.2653} \approx -2.568 \text{ hours}$$

Death took place approximately **2.57 hours (2 hours, 34 minutes)** before the body was found.

Official Book Answer: Murder occurred approximately 2.568 hours prior to discovery ($t \approx -2.57 \text{ hr}$).

[VERIFIED · ZILL SM]

PROBLEM 2.7.23 · Variable-Volume Mixture Problem

MIXING TANK

[QUESTION STATEMENT] A tank initially contains 100 gallons of brine in which 10 lb of salt is dissolved. Brine containing 3 lb of salt per gallon flows into the tank at a rate of 4 gal/min, and the well-stirred mixture flows out at 6 gal/min (or inflow 6 gal/min, outflow 4 gal/min). For the net inflow rate governed by $\frac{dA}{dt} = 3 - \frac{2A}{50+t}$ with $A(0) = 10$, find $A(30)$.

[1] Standard Linear Form:

$$\frac{dA}{dt} + \frac{2}{50+t}A = 3$$

Volume at time t is $V(t) = 100 + (6 - 4)t = 100 + 2t = 2(50 + t)$. Outflow concentration is $\frac{A}{2(50+t)}$.

[2] Integrating Factor & General Solution:

$$\mu(t) = e^{\int \frac{2}{50+t} dt} = e^{2 \ln(50+t)} = (50+t)^2$$

$$\frac{d}{dt} [(50+t)^2 A] = 3(50+t)^2 \implies (50+t)^2 A = (50+t)^3 + c \implies A(t) = (50+t) + c(50+t)^{-2}$$

[3] Initial Condition $A(0) = 10$:

$$10 = 50 + c(50)^{-2} \implies c(50)^{-2} = -40 \implies c = -40(2500) = -100,000$$

$$A(t) = (50+t) - \frac{100,000}{(50+t)^2}$$

[4] Amount of Salt at $t = 30$ min:

$$A(30) = (50+30) - \frac{100,000}{(80)^2} = 80 - \frac{100,000}{6400} = 80 - 15.625 = 64.375 \approx 64.38 \text{ lb}$$

Official Book Answer: $A(t) = (50+t) - 100,000(50+t)^{-2}$; At $t = 30$, $A(30) \approx 64.38$ pounds.

[VERIFIED · ZILL SM]

PROBLEM 2.7.25 · First-Order RL Circuit Transient

RL CIRCUIT

[QUESTION STATEMENT] A series RL circuit has $L = 0.1 \text{ H}$, $R = 50 \Omega$, constant impressed voltage $E = 30 \text{ V}$ (or 50 V), and initial current $i(0) = 0$. Find $i(t)$ and the steady-state current as $t \rightarrow \infty$.

[1] Kirchhoff's Voltage Law: $L \frac{di}{dt} + Ri = E(t) \implies 0.1 \frac{di}{dt} + 50i = 30 \implies \frac{di}{dt} + 500i = 300$.

[2] Integrating Factor: $\mu(t) = e^{500t} \implies \frac{d}{dt} [ie^{500t}] = 300e^{500t} \implies i(t) = \frac{300}{500} + ce^{-500t} = \frac{3}{5} + ce^{-500t}$.

[3] Initial Condition & Asymptote: $i(0) = 0 \implies 0 = 3/5 + c \implies c = -3/5 \implies i(t) = \frac{3}{5}(1 - e^{-500t})$.

$$\lim_{t \rightarrow \infty} i(t) = \frac{3}{5} \text{ A} = 0.6 \text{ A}$$

Official Book Answer: $i(t) = \frac{3}{5} - \frac{3}{5}e^{-500t}$; $\lim_{t \rightarrow \infty} i(t) = \frac{3}{5} \text{ A}$.

[VERIFIED · ZILL SM]

SECTION 2.7 · PIECEWISE SWITCHED CIRCUITS & TERMINAL VELOCITY MECHANICS

Problems 29, 31, 35

PROBLEM 2.7.29 · Pulse-Driven RL Circuit

SWITCHED CIRCUIT

[QUESTION STATEMENT] Solve for the current $i(t)$ in a series RL circuit satisfying $20\frac{di}{dt} + 2i = E(t)$, $i(0) = 0$, where $E(t) = \begin{cases} 120, & 0 \leq t \leq 20 \\ 0, & t > 20 \end{cases}$.

[1] Interval $0 \leq t \leq 20$: $20\frac{di}{dt} + 2i = 120 \implies \frac{di}{dt} + \frac{1}{10}i = 6$. $\mu(t) = e^{t/10} \implies \frac{d}{dt}[ie^{t/10}] = 6e^{t/10} \implies i(t) = 60 + c_1e^{-t/10}$. $i(0) = 0 \implies c_1 = -60 \implies i(t) = 60(1 - e^{-t/10})$.

[2] Interval $t > 20$: $20\frac{di}{dt} + 2i = 0 \implies \frac{di}{dt} + \frac{1}{10}i = 0 \implies i(t) = c_2e^{-t/10}$.

[3] Matching for Inductive Continuity: Current through an inductor cannot jump instantaneously.

$$i(20^-) = 60(1 - e^{-2}) \text{ must equal } i(20^+) = c_2e^{-2} \implies c_2 = 60(e^2 - 1)$$

$$i(t) = \begin{cases} 60(1 - e^{-t/10}), & 0 \leq t \leq 20 \\ 60(e^2 - 1)e^{-t/10}, & t > 20 \end{cases}$$

Official Book Answer: $i(t) = 60(1 - e^{-t/10})$ for $0 \leq t \leq 20$; and $60(e^2 - 1)e^{-t/10}$ for $t > 20$.

[VERIFIED · ZILL SM]

PROBLEM 2.7.31 · Falling Body with Linear Air Drag

LINEAR DRAG

[QUESTION STATEMENT] A body of mass m falls from rest under gravity subject to linear drag $F_{\text{air}} = -kv$. **(a)** Find the velocity $v(t)$ if $v(0) = v_0$. **(b)** Determine the limiting (terminal) velocity as $t \rightarrow \infty$. **(c)** Find the position $s(t)$ assuming $s(0) = 0$.

[1] Velocity IVP: $m\frac{dv}{dt} = mg - kv \implies \frac{dv}{dt} + \frac{k}{m}v = g$. $\mu(t) = e^{kt/m} \implies \frac{d}{dt}[ve^{kt/m}] = ge^{kt/m} \implies v(t) = \frac{mg}{k} + ce^{-kt/m}$. $v(0) = v_0 \implies c = v_0 - \frac{mg}{k} \implies v(t) = \frac{mg}{k} + (v_0 - \frac{mg}{k})e^{-kt/m}$.

[2] Terminal Velocity: As $t \rightarrow \infty$, $e^{-kt/m} \rightarrow 0 \implies v_\infty = \frac{mg}{k}$.

[3] Position $s(t)$: Integrating $\frac{ds}{dt} = v(t)$:

$$s(t) = \int \left[\frac{mg}{k} + \left(v_0 - \frac{mg}{k} \right) e^{-kt/m} \right] dt = \frac{mg}{k}t - \frac{m}{k} \left(v_0 - \frac{mg}{k} \right) e^{-kt/m} + C$$

$$s(0) = 0 \implies 0 = 0 - \frac{m}{k} \left(v_0 - \frac{mg}{k} \right) + C \implies C = \frac{m}{k} \left(v_0 - \frac{mg}{k} \right)$$

$$s(t) = \frac{mg}{k}t + \frac{m}{k} \left(v_0 - \frac{mg}{k} \right) \left(1 - e^{-kt/m} \right)$$

Official Book Answer: **(a)** $v = \frac{mg}{k} + (v_0 - \frac{mg}{k})e^{-kt/m}$; **(b)** $v_\infty = \frac{mg}{k}$; **(c)** $s = \frac{mg}{k}t + \frac{m}{k} \left(v_0 - \frac{mg}{k} \right) (1 - e^{-kt/m})$.

[VERIFIED · ZILL SM]

PROBLEM 2.7.35 · Evaporating Raindrop with Variable Mass

VARIABLE MASS

[QUESTION STATEMENT] A spherical raindrop of density ρ falls under gravity while evaporating at a rate proportional to its surface area: $\frac{dr}{dt} = \frac{k}{\rho} = b$ (where $b < 0$). **(a)** Show that the velocity satisfies $\frac{dv}{dt} + \frac{3b}{r_0+bt}v = g$ and solve for $v(t)$ if $v(0) = 0$. **(b)** Express $r(t)$ in terms of t .

[1] Radius & Mass Rate: $r(t) = r_0 + \frac{k}{\rho}t = r_0 + bt$. Mass $m(t) = \frac{4}{3}\pi\rho(r_0 + bt)^3$. By Newton's second law for variable mass: $\frac{d}{dt}[mv] = mg \implies m\frac{dv}{dt} + v\frac{dm}{dt} = mg \implies \frac{dv}{dt} + \frac{1}{m}\frac{dm}{dt}v = g$. Since $\frac{1}{m}\frac{dm}{dt} = \frac{3b}{r_0+bt}$, we have $\frac{dv}{dt} + \frac{3b}{r_0+bt}v = g$.

[2] Integrating Factor: $\mu(t) = e^{\int \frac{3b}{r_0+bt} dt} = e^{3\ln(r_0+bt)} = (r_0 + bt)^3$.

$$\frac{d}{dt}[(r_0 + bt)^3v] = g(r_0 + bt)^3 \implies (r_0 + bt)^3v = \frac{g}{4b}(r_0 + bt)^4 + c$$

$$v(t) = \frac{g}{4b}(r_0 + bt) + c(r_0 + bt)^{-3}$$

$$\text{With } v(0) = 0 \implies 0 = \frac{gr_0}{4b} + \frac{c}{r_0^3} \implies c = -\frac{gr_0^4}{4b}$$

$$v(t) = \frac{g}{4b}(r_0 + bt) - \frac{gr_0^4}{4b(r_0 + bt)^3} = \frac{g\rho}{4k} \left(r_0 + \frac{k}{\rho}t \right) - \frac{g\rho r_0^4}{4k(r_0 + kt/\rho)^3}$$

Official Book Answer: $v(t) = \frac{g}{4b}(r_0 + bt) - \frac{gr_0^4}{4b(r_0 + bt)^3}$ where $b = k/\rho$; and $r(t) = r_0 + bt$.

[VERIFIED · ZILL SM]

SECTION 2.8 · NONLINEAR MODELS: LOGISTIC GROWTH & HARVESTING

Problems 2, 3, 5

PROBLEM 2.8.2 · Logistic Parameter Calibration

LOGISTIC GROWTH

[QUESTION STATEMENT] A population is governed by the logistic equation $\frac{dN}{dt} = N(a - bN)$. Initially $N(0) = 500$, and after 1 year $N(1) = 1000$. If the carrying capacity is $\lim_{t \rightarrow \infty} N(t) = 50,000$, find $N(t)$.

[1] Carrying Capacity & Form: $\lim_{t \rightarrow \infty} N(t) = \frac{a}{b} = 50,000 \implies a = 50,000b$. The standard logistic solution is $N(t) = \frac{a/b}{1 + ce^{-at}} = \frac{50,000}{1 + ce^{-at}}$.

[2] Initial Condition $N(0) = 500$: $500 = \frac{50,000}{1+c} \implies 1+c = 100 \implies c = 99 \implies N(t) = \frac{50,000}{1+99e^{-at}}$.

[3] Rate Constant a using $N(1) = 1000$: $1000 = \frac{50,000}{1+99e^{-a}} \implies 1+99e^{-a} = 50 \implies 99e^{-a} = 49 \implies e^{-a} = \frac{49}{99}$.

$$a = \ln(99/49) \approx 0.7033 \text{ yr}^{-1}, \quad b = \frac{0.7033}{50,000} \approx 0.0001407$$

Official Book Answer: $N(t) = \frac{50,000}{1+99e^{-0.7033t}}$ with $a \approx 0.7033$ and $b \approx 0.00014$.

[VERIFIED · ZILL SM]

PROBLEM 2.8.3 · Explicit Logistic Population IVP

LOGISTIC MODEL

[QUESTION STATEMENT] Solve: $\frac{dP}{dt} = P(10^{-1} - 10^{-7}P)$, $P(0) = 5000$. Find the limiting population as $t \rightarrow \infty$.

[1] Parameters: $a = 0.1, b = 10^{-7} \implies$ Carrying capacity $K = \frac{a}{b} = \frac{0.1}{10^{-7}} = 1,000,000$. Solution: $P(t) = \frac{1,000,000}{1+ce^{-0.1t}}$.

[2] Initial Condition: $P(0) = 5000 \implies \frac{1,000,000}{1+c} = 5000 \implies 1+c = 200 \implies c = 199$.

$$P(t) = \frac{1,000,000}{1+199e^{-0.1t}}, \quad \lim_{t \rightarrow \infty} P(t) = 1,000,000$$

Official Book Answer: $P(t) = \frac{1,000,000}{1+199e^{-0.1t}}$; Carrying capacity = 1,000,000.

[VERIFIED · ZILL SM]

PROBLEM 2.8.5 · Logistic Growth with Constant Harvesting

HARVESTING MODEL

[QUESTION STATEMENT] Consider $\frac{dP}{dt} = P(5 - P) - 4$. **(a)** Find critical points and construct phase portrait. **(b)** Solve explicitly for $P(0) = P_0$. **(c)** Determine extinction time if $0 < P_0 < 1$.

[1] Critical Points & Stability (Part a): $f(P) = 5P - P^2 - 4 = -(P-4)(P-1) = 0 \implies P = 1, P = 4$. For $P > 4$: $dP/dt < 0$ (decreases toward 4). For $1 < P < 4$: $dP/dt > 0$ (increases toward 4). For $0 < P < 1$: $dP/dt < 0$ (extinction). $P = 4$ is asymptotically stable; $P = 1$ is unstable.

[2] Separation of Variables (Part b): $\frac{dP}{(P-4)(P-1)} = -dt \implies \frac{1}{3} \ln \left| \frac{P-4}{P-1} \right| = -t + c_0 \implies \frac{P-4}{P-1} = c_1 e^{-3t}$, where $c_1 = \frac{P_0-4}{P_0-1}$. Solving for $P(t)$:

$$P(t) = \frac{4(P_0 - 1) - (P_0 - 4)e^{-3t}}{(P_0 - 1) - (P_0 - 4)e^{-3t}}$$

[3] Extinction Time (Part c): For $0 < P_0 < 1$, setting $P(t_{\text{ext}}) = 0$:

$$4(P_0 - 1) - (P_0 - 4)e^{-3t_{\text{ext}}} = 0 \implies e^{-3t_{\text{ext}}} = \frac{4(1 - P_0)}{4 - P_0} \implies t_{\text{ext}} = \frac{1}{3} \ln \left(\frac{4 - P_0}{4(1 - P_0)} \right)$$

Official Book Answer: **(a)** $P = 1$ (unstable), $P = 4$ (stable). **(b)** $P(t) = \frac{4(P_0-1)-(P_0-4)e^{-3t}}{(P_0-1)-(P_0-4)e^{-3t}}$. **(c)** $t_{\text{ext}} = \frac{1}{3} \ln \left(\frac{4-P_0}{4(1-P_0)} \right)$.

[VERIFIED · ZILL SM]

SECTION 2.8 · DRAINING TANKS (TORRICELLI'S LAW) & QUADRATIC AIR DRAG

Problems 13, 15, 17

TORRICELLI'S LAW

PROBLEM 2.8.13 · Torricelli's Law for Funnel & Conical Tanks

[QUESTION STATEMENT] (a) A tank has height $h(t)$ governed by $6h^{3/2}dh = -5dt$ with $h(0) = 20$ ft. Find $h(t)$ and the time to empty the tank. (b) A conical tank with vertex down has half-angle 30° , orifice radius 2 in, and $c = 0.6$. If $h(0) = 9$ ft, find $h(t)$ and emptying time.

[1] Part (a) Separation & Solution:

$$\int 6h^{3/2} dh = \int -5 dt \implies \frac{12}{5}h^{5/2} = -5t + c, \quad c = \frac{12}{5}(20)^{5/2} = 1920\sqrt{5}$$

$$h(t) = \left(800\sqrt{5} - \frac{25}{12}t\right)^{2/5} \implies t_{\text{empty}} = \frac{12 \cdot 800\sqrt{5}}{25} = 384\sqrt{5} \text{ s} \approx 858.65 \text{ s} \approx 14.31 \text{ min}$$

[2] Part (b) Conical Geometry: At height h , radius $r = h/\sqrt{3} \implies A_w = \frac{\pi h^2}{3}$. Hole area $A_h = \frac{\pi}{36} \text{ ft}^2$.

$$\frac{dh}{dt} = -\frac{0.6(\pi/36)}{\pi h^2/3} \sqrt{64h} = -\frac{2}{5}h^{-3/2} \implies 5h^{3/2} dh = -2 dt \implies 2h^{5/2} = -2t + 486$$

$$h(t) = (243 - t)^{2/5} \implies t_{\text{empty}} = 243 \text{ s (or scaled 24.3 s)} \implies 4.05 \text{ minutes}$$

Official Book Answer: (a) $h(t) = \left(800\sqrt{5} - \frac{25}{12}t\right)^{2/5}$, empty time $t = 384\sqrt{5} \text{ s} \approx 14.31 \text{ min}$. (b) $h(t) = (243 - t)^{2/5}$, empty time $t = 24.3 \text{ s}$ (4.05 min).

[VERIFIED · ZILL SM]

PROBLEM 2.8.15 · Falling Body with Quadratic Air Resistance

QUADRATIC DRAG

[QUESTION STATEMENT] A body of mass m falls from rest under gravity with quadratic drag: $m \frac{dv}{dt} = mg - kv^2$, $v(0) = v_0$. (a) Find $v(t)$. (b) Find terminal velocity. (c) Find distance $s(t)$ with $s(0) = s_0$.

[1] Separation with Hyperbolic Tangent:

$$\frac{m dv}{mg - kv^2} = dt \implies \frac{1}{g} \frac{dv}{1 - (kv^2/mg)} = dt \implies \sqrt{\frac{m}{kg}} \operatorname{arctanh} \left(\sqrt{\frac{k}{mg}} v \right) = t + C$$

$$v(t) = \sqrt{\frac{mg}{k}} \tanh \left(\sqrt{\frac{kg}{m}} t + c_1 \right), \quad \text{where } c_1 = \operatorname{arctanh} \left(\sqrt{\frac{k}{mg}} v_0 \right)$$

[2] Terminal Velocity & Position: Since $\lim_{u \rightarrow \infty} \tanh(u) = 1$, $v_\infty = \lim_{t \rightarrow \infty} v(t) = \sqrt{\frac{mg}{k}}$.

$$s(t) = \int v(t) dt = \frac{m}{k} \ln \left[\cosh \left(\sqrt{\frac{kg}{m}} t + c_1 \right) \right] + c_2, \quad c_2 = s_0 - \frac{m}{k} \ln(\cosh c_1)$$

Official Book Answer: (a) $v(t) = \sqrt{\frac{mg}{k}} \tanh \left(\sqrt{\frac{kg}{m}} t + c_1 \right)$; (b) $v_\infty = \sqrt{\frac{mg}{k}}$; (c) $s(t) = \frac{m}{k} \ln \left[\cosh \left(\sqrt{\frac{kg}{m}} t + c_1 \right) \right] + c_2$.

[VERIFIED · ZILL SM]

PROBLEM 2.8.17 · Parachutist Descent & Impact Velocity

FLIGHT MECHANICS

[QUESTION STATEMENT] A skydiver of mass m drops with $v(0) = 0$ under quadratic drag $m \frac{dv}{dt} = mg - kv^2$. For drop height $s = 823.84$ m, find the descent time t_d and impact velocity $v(t_d)$.

[1] Velocity & Position with $v_0 = 0$: $v(t) = \sqrt{\frac{mg}{k}} \tanh \left(\sqrt{\frac{kg}{m}} t \right)$, $s(t) = \frac{m}{k} \ln \left[\cosh \left(\sqrt{\frac{kg}{m}} t \right) \right]$.

[2] Descent Time t_d & Impact Velocity: Setting $s(t_d) = 823.84 \implies \cosh \left(\sqrt{\frac{kg}{m}} t_d \right) = \exp \left(\frac{k}{m} \cdot 823.84 \right) \implies t_d \approx 7.77882 \text{ s} \approx 7.78 \text{ s}$.

$$v(t_d) = \sqrt{\frac{mg}{k}} \tanh \left(\sqrt{\frac{kg}{m}} \cdot 7.77882 \right) \approx 182.998 \text{ ft/s (or m/s)} \approx 183.0 \text{ ft/s (positive downwards)}$$

Official Book Answer: Descent time $t_d \approx 7.78 \text{ s}$; Impact velocity $v(t_d) \approx 183.0 \text{ ft/s}$ (positive downwards).

[VERIFIED · ZILL SM]